

SCI-BEAM

Scientific Rationale and Contract

Implementation-blind Stage B draft
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Draft status. Owner-authorized scientific draft, not accepted or frozen. It states algebraic authority only. It makes no claim of implementation conformance, representation/response fidelity, observational performance, or production readiness.

Compact contract opening

Inputs	Immutable observation/calibrator identity; declared source brightness/flux model; detector-resolved conditioned signal or complete map bundle; WCS-bound relative coordinates; explicit detector metadata; covariance or typed approximation; requested/effective/realized policy; optional compatible soft prior.
Outputs	Atomic per-observation bundle containing every detector attempt and terminal state, model/parameter estimates, translated source-reference centroid and shape, reference-origin excess amplitude, uncertainty status, prior influence, convergence/QC causes, typed calibration-candidate state, and compatibility provenance.
Equation	$y_i = AT(\mathbf{x}_i - \boldsymbol{\mu}; \boldsymbol{\Sigma}, \boldsymbol{\phi}) + \mathbf{b}_i^T \boldsymbol{\beta} + r_i$, fitted by declared generalized least squares or likelihood on causally admitted support.
Source	First-principles derivation under the owner-approved SCI-BEAM Scope Brief and sanitized conventions. Bryan et al. (2018) and Golec et al. (2024) supply bounded instrument context only.
Status	v0.1/r0.1 draft. Numerical production thresholds, calibration promotion, sensitivity authority, upstream closure, and current implementation behavior are intentionally unavailable.

1 Scientific question and boundary

SCI-BEAM asks a narrow question: given an admitted observation of a declared calibrator and a trustworthy relation between retained data and a relative sky plane, what observation-local source-convolved profile is supported for each detector, with what uncertainty and validity? This question is smaller than astrometry, absolute calibration, sensitivity, mapmaking, and recurrent feedback. Keeping it smaller prevents a fitted centroid from becoming pointing truth, a fitted amplitude from becoming source flux, or a finite optimizer result from becoming production evidence.

The observation and source identities are immutable. Every detector is attempted through an explicit occurrence-scoped binding. The scientific result is the parameterized response on admitted support, not a table row or filename. BEAM consumes upstream meanings without upgrading them: an incomplete response yields an effective-profile estimate, not a complete physical beam; a provisional unit yields a coefficient in that unit, not a promoted calibration. Likewise, the fitted translation of the declared source-reference origin is a relative modeled centroid in the supplied tangent plane, not a correction to the telescope pointing model or an assertion of absolute astrometric truth.

This boundary determines what counts as an estimand. The observation-local coefficient, centroid translation, covariance ellipse, and background are quantities of one realized forward model. They exist only with their source identity, support, response state, covariance choice, and effective policy. Removing those qualifiers does not simplify the result;

it changes the question. A consumer that needs a detector-plus-telescope beam, a pointing update, or a calibrated flux must establish the additional authority rather than infer it from a plausible number.

2 Why this forward model

The minimal two-dimensional beam model must represent unequal principal widths while avoiding redundant labels. A symmetric positive-definite covariance ellipse supplies that geometry. Ordering the eigenvalues and making angle periodic over π removes the axis-swap degeneracy. At circularity the profile is rotation invariant, so reporting an angle would attach meaning to an unidentifiable coordinate.

A calibrator is not always a mathematical point. Convolution a separately declared, unit-integral brightness shape with the Gaussian separates source extent from the beam parameterization. In v0.1 the source shape also has a finite first moment and a declared reference origin at zero first moment. A centered Gaussian has the same reference centroid, so their convolution retains it. The fitted μ_d is the translation of that reference origin into the supplied tangent plane.

Equation 5 divides by the positive convolved value $H(\mathbf{0})$ and therefore establishes exactly $T(\mathbf{0}) = 1$. It does not establish that H peaks at zero: a non-negative source can be asymmetric even when its first moment is zero. The amplitude A_d is consequently the modeled source excess above background at $\mathbf{x} = \mu_d$, not necessarily the maximum value of the convolved profile. This reference-origin convention makes the coefficient reproducible across point-like and finite-source cases without silently imposing source symmetry. A later claim that A_d is a peak would require the realized source model to establish that additional property.

The reference convention also prevents the translation and amplitude from absorbing arbitrary choices in the supplied source model. If the same brightness shape were tabulated about a shifted origin, a compensating change in μ_d could leave the modeled pixels unchanged while changing the reported centroid. Fixing the first moment removes that bookkeeping degeneracy. Binding the normalization identity to every amplitude similarly prevents two otherwise identical fits, one normalized at a reference origin and another at a maximum or by integral, from masquerading as compatible measurements.

Finite-source modeling does not by itself make the intrinsic beam identifiable. For a Gaussian source and Gaussian beam, for example, observed second moments combine source and beam contributions; uncertainty in one propagates directly into the other. With a general declared source shape, the same issue appears through correlated derivatives of T with respect to Σ and ϕ . When source extent approaches the beam scale, a broad range of beam widths may be supported after plausible changes in the source nuisance parameters. The contract therefore requires the supplied source uncertainty to be propagated, or the intrinsic-width interpretation to become unavailable. Holding an uncertain source fixed can define a conditional fit, but it cannot create information.

The point-source limit is a continuity check rather than permission to ignore finite extent. As the admitted source shape tends to a delta distribution at the declared reference origin, the reference-normalized convolution tends to the centered Gaussian. Away from that limit, the realized source identity and normalization remain part of the estimand. A fit cannot select whichever source extent makes the beam look most convenient unless such selection, its identifiability, and its evidence are separately authorized.

A constant or planar background is the largest v0.1 family admitted. It can absorb a local offset or first-order slope while keeping the source model auditable. More flexible backgrounds can absorb broad source modes and require a new identifiability argument. Even the planar family is conditional: cropped or one-sided support may make gradients inseparable from centroid or width.

3 Normative notation

Symbol	Meaning
o, d, i	Observation, stable detector identity, and admitted sample or map-pixel index.
$\mathbf{x}_i = (x_i, y_i)^\top$	Indices in memory are zero-based; persisted FITS/WCS pixels are one-based. Externally supplied AltAz tangent-plane azimuth/elevation offset, in arcseconds, about the declared source. WCS supplies signs and handedness.

y_i	Admitted detector datum in declared signal unit U_y ; the active map boundary is mJy/beam but that label alone proves neither absolute calibration nor complete response.
v_i	Upstream eligibility and validity state, evaluated before the numerical payload.
$\mathcal{I}_d$	Admitted support for detector d : indices satisfying upstream eligibility, finite-data policy, coordinate validity, and the effective support rule.
$\mathbf{C}_d$	Declared covariance of $\mathbf{y}_d$ on $\mathcal{I}_d$, or a typed approximation $\tilde{\mathbf{C}}_d$.
$S(\mathbf{u}; \phi)$	Declared, non-negative calibrator surface-brightness shape with finite first moment, unit integral, and declared reference origin fixed by $\int \mathbf{u} S(\mathbf{u}) d^2 \mathbf{u} = \mathbf{0}$, with nuisance parameters ϕ .
$G(\mathbf{u}; \Sigma)$	Peak-normalized elliptical Gaussian beam.
$T(\mathbf{u}; \Sigma, \phi)$	Source-convolved template normalized to unity at the declared source-reference origin, $T(\mathbf{0}) = 1$, as defined in Section 5; that origin need not be the profile maximum.
A_d	Fitted source excess at the translated declared reference origin, in U_y ; it is neither necessarily the profile peak nor source flux, calibration, or sensitivity.
$\boldsymbol{\mu}_d = (\mu_{x,d}, \mu_{y,d})$	Fitted translation of the declared zero-first-moment source-reference origin in the admitted tangent plane, in arcseconds; equivalently the modeled convolved-profile centroid when the admitted moments exist.
$\sigma_{\text{maj}}, \sigma_{\text{min}}, \theta$	Gaussian major/minor standard deviations and orientation, with $\sigma_{\text{maj}} \geq \sigma_{\text{min}} > 0$ and $\theta \in [0, \pi)$.
$\mathbf{q}_d$	Model parameter vector $(A_d, \boldsymbol{\mu}_d, \sigma_{\text{maj}}, \sigma_{\text{min}}, \theta, \boldsymbol{\beta}_d)$.
$\boldsymbol{\beta}_d$	Coefficients of the declared bounded background basis.
Q_d	Declared generalized least-squares or likelihood objective.
$d_j(a, b)$	Declared stability metric for available parameter j ; for orientation, $d_\pi(a, b) = \min_{n \in \mathbb{Z}} a - b + n\pi $.
$\mathcal{A}_d^{(k)}, c_d^{(k)}, \mathcal{V}^{(k)}$	Parameter-availability set, selected locator/measurement-candidate identity (including a typed none state), and valid-detector set at iteration k .
$F_{s,a}$	Immutable source-model flux quantity supplied for array a , with its epoch, passband, convention, and uncertainty.
K_d	Typed candidate conversion factor $F_{s,a}/A_d$ when dimensional and authority preconditions hold.

The words *shall* and *shall not* are normative. *May* marks an allowed option whose realized state must be recorded. “Unavailable” is a typed scientific state, never an undocumented numeric sentinel.

4 Admission, support, and objective

Validity precedes arithmetic. A non-finite value multiplied by zero is not removed, and a disabled or invalid sample is not restored because its payload is finite. After upstream exclusions, the effective geometric/support policy constructs $\mathcal{I}_d$ and records its shape, connectedness, edge relation, masked core, and causes. Zero support produces no fit.

Support is part of the operator because it determines which distinctions the data can make. Symmetric coverage around the reference origin may separate a source translation from a planar background gradient, while one-sided or cropped coverage can make them nearly interchangeable. Removing a masked core may eliminate direct leverage on the reference-origin amplitude even when the wings constrain a width. A disconnected set can carry information only under an explicit rule that preserves its geometry. These are not merely data-volume effects: two supports with equal sample count can have different rank, covariance, and susceptibility to source/background degeneracy.

The objective is a declared generalized least-squares likelihood. Full covariance is scientifically preferable when correlations are material, but v0.1 permits an explicitly labeled approximation so that the point estimate and its statistical limitations can be separated. A diagonal approximation may still yield a useful optimum, yet its inverse-Hessian errors are only conditional. Silent covariance regularization would change the estimator, so any retained-subspace factorization is part of realized provenance.

Covariance has two roles that must not be conflated. In the objective it specifies how residual patterns are weighted, so changing it can move the optimum. In curvature or repeated-sampling interpretation it states how parameter

variation is calibrated, so an approximation can also change claimed precision even when the optimum barely moves. A singular covariance contains no ordinary inverse on the full sample space; retaining a subspace or regularizing modes is therefore a scientific choice about which residual combinations are measured. v0.1 permits only a declared, provenance-bearing procedure and supplies no universal regularizer.

Identifiability is local to the realized model and support. Width, source extent, background, and centroid can trade off, especially under cropping or uncertain finite-source size. A successful optimizer cannot supply information absent from the data. The contract therefore permits partial availability: a coefficient may exist while an intrinsic-beam interpretation or calibrated covariance remains unavailable.

Local curvature is useful only after that distinction is made. An inverse Hessian around an interior, identifiable optimum describes uncertainty conditional on the realized likelihood and nuisance treatment. It is not automatically a total covariance, nor does finite matrix output prove regularity. Boundaries, near-circular ellipses, weak source/background separation, or an unconstrained source extent can make some parameter directions singular or coordinate-dependent. The result may retain an identifiable invariant such as the covariance ellipse or common width while marking orientation or a deconvolved width unavailable.

5 Forward model and objective

Let

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}, \quad (1)$$

$$\mathbf{\Sigma} = \mathbf{R}(\theta) \text{diag}(\sigma_{\text{maj}}^2, \sigma_{\text{min}}^2) \mathbf{R}(\theta)^\top, \quad (2)$$

$$G(\mathbf{u}; \mathbf{\Sigma}) = \exp\left(-\frac{1}{2} \mathbf{u}^\top \mathbf{\Sigma}^{-1} \mathbf{u}\right), \quad (3)$$

$$H(\mathbf{u}; \mathbf{\Sigma}, \phi) = \int S(\mathbf{t}; \phi) G(\mathbf{u} - \mathbf{t}; \mathbf{\Sigma}) d^2 \mathbf{t}, \quad (4)$$

$$T(\mathbf{u}; \mathbf{\Sigma}, \phi) = \frac{H(\mathbf{u}; \mathbf{\Sigma}, \phi)}{H(\mathbf{0}; \mathbf{\Sigma}, \phi)}. \quad (5)$$

The admitted source model shall have finite first moment and satisfy

$$\int S(\mathbf{u}; \phi) d^2 \mathbf{u} = 1, \quad \int \mathbf{u} S(\mathbf{u}; \phi) d^2 \mathbf{u} = \mathbf{0}, \quad H(\mathbf{0}) > 0.$$

Thus the declared reference origin is the source centroid, convolution with the centered Gaussian preserves that centroid, and Equation 5 guarantees $T(\mathbf{0}) = 1$. It does not assert that $\mathbf{0}$ is the global maximum for an asymmetric source. The detector model is

$$m_i(\mathbf{q}_d, \phi) = A_d T(\mathbf{x}_i - \boldsymbol{\mu}_d; \mathbf{\Sigma}_d, \phi) + \mathbf{b}_i^\top \boldsymbol{\beta}_d, \quad (6)$$

where $\mathbf{b}_i$ is the effective constant or planar background basis.

For an admitted covariance, the estimator minimizes

$$Q_d = (\mathbf{y}_d - \mathbf{m}_d)^\top \mathbf{C}_d^{-1} (\mathbf{y}_d - \mathbf{m}_d) + \log |\mathbf{C}_d|, \quad (7)$$

up to additive constants. If only a declared diagonal or other approximation is supplied, the same expression uses $\tilde{\mathbf{C}}_d$ and every covariance, standardized-residual, and significance claim is labeled conditional on that approximation. Singular covariance is handled by a declared, provenance-bearing factorization on an explicitly retained subspace; silent regularization is forbidden.

The fit covariance may be reported as a local likelihood covariance

$$\mathbf{V}_{q,d} = \left[\frac{1}{2} \nabla \nabla^\top Q_d(\hat{\mathbf{q}}_d) \right]^{-1} \quad (8)$$

only when the retained parameterization is locally identifiable, the solution is interior for the reported parameters, and the likelihood/covariance assumptions used by the formula are declared. Otherwise it is unavailable or supplied by a separately identified valid procedure. With nuisance vector $\boldsymbol{\eta}$ and admitted covariance $\mathbf{V}_\eta$, first-order propagation is

$$\mathbf{V}_{\text{total}} = \mathbf{V}_q + \mathbf{J}_\eta \mathbf{V}_\eta \mathbf{J}_\eta^\top \quad (9)$$

with cross-terms included when dependence exists. Omitted calibrator, coordinate, response, atmosphere, bandpass, or calibration uncertainty is named; “total” is prohibited when a material term is unavailable.

When the admitted source flux $F_{s,a}$ and fitted coefficient A_d have compatible conventions and $A_d \neq 0$, BEAM may form

$$K_d = \frac{F_{s,a}}{A_d}, \quad \text{Var}(K_d) \simeq \frac{\text{Var}(F_{s,a})}{A_d^2} + \frac{F_{s,a}^2 \text{Var}(A_d)}{A_d^4} - 2 \frac{F_{s,a} \text{Cov}(F_{s,a}, A_d)}{A_d^3}. \quad (10)$$

If any required variance or dependence term is unknown, candidate uncertainty is unavailable rather than silently independent. K_d remains a typed candidate for SCI-CAL review.

An observation-local iteration may alternate locator and measurement phases. The effective policy declares a metric d_j and scale s_j for each parameter, objective scale s_Q , tolerances ϵ_j, ϵ_Q , support and valid-detector stability rules, and maximum iteration $N_{\max}$. Linear scalar parameters may use absolute difference only when the policy says so; orientation shall use the modulo- π distance

$$d_\pi(a, b) = \min_{n \in \mathbb{Z}} |a - b + n\pi|.$$

Let $\mathcal{A}_d^{(k)}$ be the parameter-availability set, $c_d^{(k)}$ the selected locator/measurement-candidate identity including a typed none state, and $\mathcal{V}^{(k)}$ the valid-detector set. Convergence occurs only when

$$\begin{aligned} \max_{j \in \mathcal{A}_d^{(k)} \cap \mathcal{A}_d^{(k-1)}} \frac{d_j(q_j^{(k)}, q_j^{(k-1)})}{s_j} &\leq \epsilon_j, & \mathcal{A}_d^{(k)} &= \mathcal{A}_d^{(k-1)}, \\ \frac{|Q^{(k)} - Q^{(k-1)}|}{s_Q} &\leq \epsilon_Q, & c_d^{(k)} &= c_d^{(k-1)}, \\ \mathcal{I}_d^{(k)} &= \mathcal{I}_d^{(k-1)}, & \mathcal{V}^{(k)} &= \mathcal{V}^{(k-1)}. \end{aligned} \quad (11)$$

for the declared number of consecutive transitions and only where the objective is mutually defined. Availability equality is tested separately: a parameter unavailable at both transitions, including θ at circularity, contributes no numerical metric; a change into or out of availability fails stability. Candidate, support, and valid-detector stability are separately recorded components, not entries hidden in a raw parameter vector. No numerical tolerance or count is a universal scientific constant in v0.1.

6 Priors, iteration, and convergence

The admitted TolAPT prior is intentionally soft and coarser than detector truth. It may seed candidate location and bound a prior-guided search, but it is not a penalty in the measurement objective and cannot veto blind search. Candidate comparison under one measurement objective gives observed evidence the final say among admitted candidates. Route, displacement, gate, reliability, compatibility, and fallback are retained so a result can reveal prior influence.

This separation is causal, not cosmetic. A locator chooses which measurement hypotheses receive evaluation; a measurement objective scores the admitted data under the realized source-plus-background model. If prior reliability were inserted as an unreported objective penalty, a candidate could win because of inherited expectation rather than the observation. Conversely, a bounded prior-guided search alone could exclude a stronger source outside its gate. Maintaining a declared blind route and comparing candidate identities under the same measurement objective makes those two influences observable. The prior can reduce work or focus a search, but the record still shows whether it changed the searched domain or selected basin.

A prior also cannot repair incompatible identity. Array, network, slot, frame, unit, producer version, and artifact relation must agree before its location or scale is used. Repeated slot text is not detector truth, and a finite coordinate is not evidence of compatibility. Missing, weak, incompatible, or unsuccessful prior use leads to the declared blind fallback with a typed cause. This preserves reproducibility: rerunning an identical observation with the same effective policy cannot inherit an unstated candidate from an earlier observation.

Locator and measurement phases may alternate within the one observation-local estimator. That loop changes neither the immutable source nor external calibration authority. Convergence is a conjunction of separately represented stability statements under versioned metrics, scales, tolerances, and a declared consecutive-transition count. Alternation, incomplete stability, and maximum-iteration termination are distinct. There are no universal numerical defaults in this draft because selecting them is an empirical production decision.

A raw parameter-vector difference is not a scientific convergence metric. Orientation is periodic over π , so angles just below π and just above zero can represent nearly identical ellipses despite a large raw subtraction. The declared metric therefore uses modulo- π angular distance. At circularity, orientation is unavailable rather than zero; no angular difference is computed when it is unavailable at both transitions, while a change into or out of availability fails stability. Other parameters use declared metrics and scales appropriate to their identities rather than acquiring an implicit common unit.

The scalar comparisons are only one component. The parameter-availability set, measurement objective, selected candidate identity, admitted support, and valid-detector set each have an independent stability record. A loop that repeats nearly equal numbers while alternating source candidates has not converged. Nor has one whose objective is stable only because samples or detectors were dropped, or whose angle disappears as the fit crosses circularity. Convergence requires the complete conjunction for the declared number of consecutive transitions. Optimizer completion records what the numerical routine did; it does not override a failed scientific stability component.

7 Uncertainty and interpretation

Fit covariance is not a synonym for total uncertainty. Local curvature is meaningful only under the realized likelihood, identifiable interior parameterization, and adequate covariance statement. Calibrator extent/flux, coordinates, response, bandpass, atmosphere, and calibration can contribute nuisance uncertainty. Terms not supplied are named, and “total” is withheld. At circularity, uncertainty belongs to an invariant shape representation; angle uncertainty is unavailable.

Nuisance propagation must follow dependence. Source flux and fitted reference-origin amplitude can share calibration or atmospheric information, and source extent can covary strongly with fitted width. Adding marginal variances while dropping cross-terms is not conservative in a universal direction: the omitted covariance can increase or decrease the result. Equation 9 therefore stands for a declared joint propagation, while Equation 10 exposes the dependence term in the calibration candidate. When a material term or relationship is unknown, the corresponding total or candidate uncertainty is unavailable rather than silently independent.

The response boundary is equally important. Filtering, map estimation, and kernels can reshape a source. Unless their complete realized transfer is established, the fitted T describes the admitted conditioned response. This is still a legitimate estimand, but it is not silently relabeled as an intrinsic detector-plus-telescope beam.

Response completeness also limits comparisons. Two observations can return equal fitted widths while referring to different filtering or map kernels; numerical equality does not make them measurements of the same physical object. Conversely, a well-characterized conditioned response may support precise repeatability claims without supporting an intrinsic-beam claim. The result therefore carries response identity and completeness beside the parameters. Deconvolution, if later authorized by the response owner, would be a separate interpretation with its own uncertainty and validity, not an automatic consequence of this fit.

8 QC, calibration candidacy, and products

QC records causes, not a magic integer. Algebraic conditions such as nonempty support, valid normalization, identified reported parameters, and physical width ordering can determine validity. Low S/N, unusual width, large residual structure, edge proximity, or prior displacement may be review cues whose thresholds require owner-controlled policy and evidence. Model inadequacy is not optimizer failure.

The distinction between validity and review policy protects both physics and operations. A non-positive width or undefined reference normalization violates the model domain regardless of a production threshold. By contrast, deciding how much edge proximity, residual structure, or prior displacement is acceptable depends on observational performance and intended use. Encoding such cuts as if they followed from the Gaussian equations would promote an empirical operating choice into a physical constant. v0.1 therefore requires every numerical threshold to arrive as a versioned policy input with requested, effective, and realized values, while preserving the underlying quantitative diagnostic and every triggered cause.

The same rule applies to signal-to-noise language. A ratio of A_d to a conditional standard error is meaningful only with its numerator, denominator, covariance assumptions, and availability state. It becomes an empirically calibrated significance only after appropriate repeated-data evidence. A high formal ratio cannot rescue a misspecified source

family, an incomplete response, or invalid support. A low ratio may be an owner-defined review cue without making an otherwise well-defined conditional coefficient algebraically nonexistent.

When source flux and amplitude conventions are compatible, their ratio can be a typed detector-calibration candidate. Its uncertainty requires both component uncertainties and their dependence. The candidate preserves response and provisional-unit status and awaits SCI-CAL promotion. Sensitivity additionally requires an exact noise, time, atmosphere, calibration, and bandwidth convention, so it is not a BEAM-owned v0.1 output.

Compatibility for that ratio is stronger than dimensional cancellation. The numerator must identify the intended source quantity, epoch, passband, and spectral convention; the denominator must identify the reference-origin template normalization, signal unit, response state, detector, and observation. A ratio formed from a peak-normalized amplitude is not interchangeable with one formed from a reference-origin amplitude unless the realized source model proves those locations coincide. BEAM can expose the typed candidate and the reason its uncertainty is available or unavailable, but only SCI-CAL can decide whether it becomes a calibration product.

The result is published atomically. Fit, QC, identity, policy, uncertainty, prior, convergence, candidate, and provenance are one scientific commitment. Optional APT, map, or TOD representations refer to that parent; no optional artifact repairs a missing required companion. Observation combination requires explicit compatibility and never rewrites the original local result.

This atomicity prevents selective promotion. A parameter table without its support and response state is not a smaller equivalent of the result, and a successful optional map does not repair a failed provenance record. Downstream APT or recurrent use may consume compatible quantities, but it must preserve the parent identity, unavailable claims, and observation-local nature of the estimate. Later evidence can qualify or supersede an interpretation through a new governed artifact; it does not retroactively turn the original r0.1 draft result into an accepted calibration, sensitivity, or production-ready beam.

9 Definitions and state model

Estimand. For each attempted detector, SCI-BEAM estimates an observation-local, source-convolved response profile over admitted support. The base family is one elliptical Gaussian beam convolved with the separately declared calibrator brightness model, plus a declared constant or first-order planar background. The point-source case is the limiting distribution $S \rightarrow \delta$. More complex beam components are outside v0.1 unless a successor contract names their identifiability and normalization.

Ellipse convention. $+x$ is the WCS-declared azimuth-offset direction and $+y$ the elevation-offset direction. The orientation θ is measured counterclockwise in that coordinate plane from $+x$ to the major-axis eigenvector, modulo π . Major/minor ordering removes axis-label degeneracy. At exact circularity, width remains defined and θ is unavailable, not zero.

Reference origin, centroid, and amplitude convention. The admitted source shape has unit integral and finite first moment, and its declared reference origin is fixed by zero first moment. Because the Gaussian is also centered, convolution preserves that reference centroid. The fitted μ_d translates this declared origin into the admitted tangent plane; it is a relative modeled centroid, not absolute astrometry. The template is normalized by its positive value at that origin, so $T(\mathbf{0}) = 1$ exactly. Consequently A_d is the modeled source excess above background at $\mathbf{x} = \mu_d$. Neither H nor T is required to attain its global maximum there for an asymmetric admitted source. A_d is not the intrinsic source flux, unconvolved beam peak, integrated flux, response coefficient, promoted calibration, or sensitivity.

Background. The effective policy chooses either a constant basis [1] or a plane $[1, x - x_*, y - y_*]$ about a recorded origin $\mathbf{x}_*$. No higher-order or adaptive term is part of v0.1. A background coefficient is reportable only when the admitted support makes the joint source/background design identifiable.

Attempt and terminal states. Every detector bound to the observation has an attempt record. The ordered processing states are `not-attempted`, `candidate-located`, `fitted`, and `optimizer-complete`; these do not imply one another when a prior stage fails. Exactly one terminal disposition is recorded: `valid`, `rejected`, `invalid-input`,

failed, non-converged, or unavailable, with one or more cause codes. Missing, disabled, unsupported, and unavailable inputs remain distinguishable causes.

Authority layers. Algebraic contract correctness means the equations and identities are internally satisfied. Implementation conformance means an implementation realizes those equations and states. Representation/response fidelity means the admitted coordinates, support, transfer response, source model, and covariance represent the intended observation. Observational performance means bias and uncertainty have been measured on appropriate data. Production readiness additionally requires owner-approved thresholds, operations, and evidence. No layer implies a later layer.

Atomic result. One observation-local publication transaction binds the immutable observation/source identity, requested/effective/realized policy, detector identity relation, fit state and parameters, covariance statement, prior route and influence, convergence trace summary, QC causes, calibration-candidate state, and provenance. Optional map or TOD companions are views with explicit parent identity; they cannot replace the required result/QC/provenance records.

Compatibility. Two artifacts are compatible only after explicit comparison of contract version, model/normalization identity, observation and source identity, detector binding scope, frame/WCS identity, units, response status, source-model convention, covariance status, and validity. Equal filenames, row counts, row order, local UID text, or finite payloads are insufficient.

10 Assumptions, dependencies, and unavailable claims

- A1.** The immutable source identity and per-array brightness/flux model are upstream inputs. Source selection and catalog estimation are not BEAM results.
- A2.** Coordinates and WCS are externally supplied in the declared AltAz tangent plane. Relative centroid inference does not establish timing, detector-coordinate truth, pointing correction, or absolute astrometry.
- A3.** Upstream validity is causal and precedes payload evaluation. BEAM adds fit-specific admission and validity without weakening upstream exclusions.
- A4.** The admitted map/signal unit and response status are preserved. Unless the complete transfer response is established upstream, fitted shape is a conditioned effective profile and an intrinsic detector-plus-telescope beam claim is unavailable.
- A5.** The calibrator model is non-negative, unit-integral, has a finite first moment with declared reference origin satisfying $\int \mathbf{u} S(\mathbf{u}) d^2\mathbf{u} = \mathbf{0}$, is evaluated for the declared epoch/passband convention, and is accompanied by uncertainty or an explicit unavailable state.
- A6.** The base beam family is adequate only as a declared model. Good residuals or convergence do not prove that adequacy; model-inadequacy diagnostics remain distinct from optimizer failure.
- A7.** The covariance or approximation describes the retained support sufficiently for the statistical claims made. A diagonal approximation does not authorize calibrated uncertainty under correlated residuals.
- A8.** The support and background basis jointly identify source, centroid, widths, angle where non-circular, amplitude, and background. Parameters that are not identified are unavailable.
- A9.** A compatible TolAPT prior is soft locator initialization and bounded gating information only. It contributes no penalty to Equation 7, is not exact UID truth, and cannot veto the declared blind route.
- A10.** Internal iteration is confined to one immutable observation/result identity. Cross-observation learning, feedback, recurrence, and restart are FRUIT responsibilities.
- A11.** SCI-CAL alone may promote a calibration candidate. Sensitivity is unavailable as a BEAM-owned result in v0.1 because no exact joint convention for noise, time, atmosphere, calibration, and bandwidth is authorized.
- A12.** Production thresholds are policy inputs, not inferred physical constants. Their provenance, requested value, effective value, realized value, owner, and change identity are recorded.

Dependencies on TolProj/TolTECA photometry, SCI-CAL, ALIGN/AST, RTC/PTC, VAL, SCI-MAP, TolAPT, `toltec_beammap`, and FRUIT remain open at their stated interfaces. This contract does not certify those inputs or close their authority decisions.

11 Evidence and promotion boundary

An analytic identity or synthetic recovery can establish algebraic correctness. Code review and tests can establish implementation conformance. Transfer-function and coordinate checks are needed for representation/response fidelity. Repeated observations and truth-bearing data are needed for bias and uncertainty performance. Production readiness further needs owner-approved numerical policies and operational evidence. These gates are deliberately non-transitive.

The predictions in Appendix B are routed accordingly. Reference-origin recovery, modulo- π invariance, blind-prior fallback, and explicit stability transitions are algebraic or state-machine tests of this contract. They can expose a mismatch between an implementation and the shared core, but even perfect synthetic recovery does not validate the upstream source model, transfer response, covariance calibration, or production thresholds. Each stronger claim needs evidence from its owning layer and must remain unavailable until that evidence is admitted.

The two admitted primary references establish only that TolTEC is a three-band millimeter-wave camera with optical context relevant to resolution/sensitivity, and that bright-source commissioning Beammap work publicly described detector location, raw-to-flux-density conversion, and beam FWHM outputs. They do not supply this estimator, thresholds, covariance, calibration meaning, or readiness evidence.

A Normative requirements

ID	Requirement
SCI-BEAM-REQ-001	The result shall bind immutable observation, source, array, detector, parent-input, contract-version, and realized-policy identities.
SCI-BEAM-REQ-002	Stable detector binding shall be explicit and occurrence-scoped; row order, slot, array, network, or local key text shall not substitute for it.
SCI-BEAM-REQ-003	Requested, effective, observation-resolved, and realized model/support/prior/iteration/QC state shall be distinct, immutable, and one-way.
SCI-BEAM-REQ-004	Coordinates shall be the supplied AltAz tangent-plane azimuth/elevation offsets in arcseconds, with WCS identity, signs, handedness, validity, and uncertainty state recorded.
SCI-BEAM-REQ-005	The source input shall declare immutable identity, epoch, passband/spectral convention, extent, flux/brightness normalization, uncertainty, and upstream authority.
SCI-BEAM-REQ-006	The signal/map input shall declare unit, response/kernel status, calibration lineage, support, validity, covariance status, provenance, and stable detector binding.
SCI-BEAM-REQ-007	Upstream-invalid samples or detectors shall be excluded before payload evaluation; missing, disabled, rejected, invalid, failed, and unavailable shall remain distinct.
SCI-BEAM-REQ-008	The v0.1 forward model shall be Equation 6: a declared finite-source or point-limit template convolved with one elliptical Gaussian plus a constant or planar background.
SCI-BEAM-REQ-009	The ellipse shall use ordered positive axes, angle measured counterclockwise from WCS $+x$, period π , and unavailable angle at exact circularity.
SCI-BEAM-REQ-010	The template shall use Equation 5; the source shall satisfy the declared zero-first-moment reference convention, and the $T(\mathbf{0}) = 1$ reference-origin normalization and source-model identity shall accompany every amplitude.
SCI-BEAM-REQ-011	A_d shall mean modeled source excess at the translated declared reference origin in the admitted signal unit; it shall not be labeled a profile peak unless separately established, nor relabeled as intrinsic flux, response, promoted calibration, or sensitivity.
SCI-BEAM-REQ-012	The background family, origin, and coefficients shall be recorded; unsupported adaptive or higher-order backgrounds shall not be introduced.
SCI-BEAM-REQ-013	Admitted support shall record geometric selection, exclusions and causes, connectedness, edge/crop state, and retained sample count.
SCI-BEAM-REQ-014	The objective shall be Equation 7 with declared covariance or typed approximation; silent weighting or regularization is forbidden.
SCI-BEAM-REQ-015	Residuals shall mean admitted data minus the complete realized source-plus-background model on the retained support.
SCI-BEAM-REQ-016	Local identifiability shall be assessed for the realized support and model; non-identifiable parameters shall be unavailable with causes.
SCI-BEAM-REQ-017	Fit covariance shall identify its method and assumptions and shall satisfy Equation 8 only where those assumptions hold.
SCI-BEAM-REQ-018	Material source, coordinate, response, bandpass, atmosphere, and calibration nuisance uncertainties shall be propagated by a declared method or named unavailable.
SCI-BEAM-REQ-019	A covariance derived under an approximation shall be labeled conditional and shall not be described as calibrated by algebra alone.
SCI-BEAM-REQ-020	Response interpretation shall preserve the upstream response/kernel state; an incomplete state shall make intrinsic-beam interpretation unavailable.
SCI-BEAM-REQ-021	A prior shall record producer/version, array/network/slot identity, frame, unit, location, scale/covariance, reliability, compatibility, and fallback state.
SCI-BEAM-REQ-022	A compatible prior may initialize and bound locator gating but shall not enter the measurement objective as exact truth or veto the blind route.

ID	Requirement
SCI-BEAM-REQ-023	Missing, incompatible, weak, or unsuccessful prior use shall invoke the declared blind locator fallback and record the selected route.
SCI-BEAM-REQ-024	Candidate comparison shall use a common realized measurement model/objective so that a stronger observed compatible source can defeat a wrong soft prior.
SCI-BEAM-REQ-025	Prior influence shall report candidate route, displacement from prior, whether gating altered the searched domain, and whether the selected candidate came from prior or blind search.
SCI-BEAM-REQ-026	Locator and measurement phases, selected-candidate identity, parameter availability, support, valid-detector set, model identity, and their changes shall be recorded as one observation-local estimator trace.
SCI-BEAM-REQ-027	Convergence shall use all separately declared components of Equation 11, including per-parameter metrics, modulo- π angular distance, availability equality, objective, candidate, support, and valid-detector stability; optimizer completion alone shall not imply convergence.
SCI-BEAM-REQ-028	Alternating candidates, changed parameter availability, violated parameter/objective/support/valid-detector stability, and maximum-iteration termination shall have distinct non-converged causes.
SCI-BEAM-REQ-029	Every attempted detector shall have an attempt record and exactly one terminal disposition with cause codes.
SCI-BEAM-REQ-030	Parameter admissibility shall require finite values where defined, positive ordered widths, a valid normalization, and parameters within the declared model domain.
SCI-BEAM-REQ-031	BEAM-specific QC shall separate quantitative validity conditions from review cues and shall preserve all triggered causes rather than collapse them into one flag.
SCI-BEAM-REQ-032	Any numerical QC, convergence, support, or S/N threshold shall be a versioned policy input with owner and requested/effective/realized values; v0.1 defines no production defaults.
SCI-BEAM-REQ-033	A signal-to-noise quantity shall name its numerator, uncertainty denominator, covariance assumptions, and status; a formal ratio shall not be called empirical significance without evidence.
SCI-BEAM-REQ-034	Model inadequacy, low S/N, edge truncation, masked core, disconnected support, nonphysical parameters, prior domination, and non-convergence shall remain distinguishable states or causes.
SCI-BEAM-REQ-035	A typed calibration candidate may be formed only by Equation 10 with compatible source/amplitude conventions and declared uncertainty/dependence status.
SCI-BEAM-REQ-036	A calibration candidate shall carry source, amplitude, response, unit, covariance, observation, detector, model, and validity identities and shall not claim SCI-CAL promotion.
SCI-BEAM-REQ-037	Sensitivity shall be unavailable as a BEAM-owned v0.1 result; carried downstream diagnostics shall preserve their external authority and convention.
SCI-BEAM-REQ-038	Publication shall be atomic across result, QC, identity, policy, uncertainty, prior, convergence, candidate, and provenance records.
SCI-BEAM-REQ-039	Failure of a required bundle member shall fail publication; an optional map or TOD companion shall not substitute for it.
SCI-BEAM-REQ-040	Optional fit, APT, map, and TOD views shall carry explicit parent and compatibility identities and preserve BEAM state and limitations.
SCI-BEAM-REQ-041	Combining observations or using a result as a later prior shall require explicit compatibility checks and shall not mutate or retroactively redefine either observation-local result.
SCI-BEAM-REQ-042	Runs shall initialize all observation-local source, flux, prior, fit, convergence, QC, and validity state; execution order shall not affect an identical observation result.
SCI-BEAM-REQ-043	Duplicate rows, permuted rows, or repeated slot labels shall be resolved only through stable binding; ambiguous binding shall reject the affected relation.
SCI-BEAM-REQ-044	Equal numerical payloads with different validity or authority metadata shall remain scientifically distinct and shall not be silently merged.
SCI-BEAM-REQ-045	Outputs shall distinguish algebraic correctness, implementation conformance, response/representation fidelity, observational performance, and production readiness.
SCI-BEAM-REQ-046	Each claim unavailable because an upstream identity, covariance, response, calibration, or validity dependency is open shall be represented explicitly and causally.

B Falsifiable predictions and edge cases

These predictions test the contract, not current software or production readiness.

ID	Predicted invariant or state
SCI-BEAM-PRED-001	Noiseless data generated by the admitted circular model on identifying support recover A as the excess at the translated declared reference origin, the zero-first-moment reference centroid, and common width; orientation is unavailable and objective residual is zero to numerical precision.
SCI-BEAM-PRED-002	Noiseless data generated by a non-circular admitted ellipse recover the translated zero-first-moment reference centroid, reference-origin amplitude, ordered widths, and θ modulo π on identifying support.
SCI-BEAM-PRED-003	Swapping unconstrained axis labels and adding $\pi/2$ leaves the profile invariant, but canonicalization restores major/minor ordering and the same canonical angle.
SCI-BEAM-PRED-004	Adding π to orientation leaves the model invariant; reported angles canonicalize to $[0, \pi)$.
SCI-BEAM-PRED-005	A constant background is recovered without changing source parameters for noiseless identifying support when the constant family is realized.
SCI-BEAM-PRED-006	A planar background is recovered under a realized planar family only when support identifies its gradients; fitting a constant to a true gradient produces structured residual/model-inadequacy evidence rather than an authorized gradient estimate.
SCI-BEAM-PRED-007	As finite source extent tends to zero while preserving the declared reference origin, the convolved $T(\mathbf{0}) = 1$ template tends to the point-source Gaussian; estimates change continuously where the problem remains identifiable.
SCI-BEAM-PRED-008	As source extent approaches beam scale, beam/source nuisance covariance grows or identifiability is lost; uncertainty shall reflect the supplied source-model uncertainty or the intrinsic-width claim becomes unavailable.
SCI-BEAM-PRED-009	Cropping or edge truncation can bias or de-identify centroid/shape; the result records edge state and cannot be valid merely because the optimizer converged.
SCI-BEAM-PRED-010	A masked core can remove reference-origin amplitude or width identifiability; those parameters become unavailable or rejected with a masked-core cause.
SCI-BEAM-PRED-011	Disconnected support is not silently joined; it is either admitted under an explicit support rule with identifiability demonstrated or rejected with cause.
SCI-BEAM-PRED-012	Zero admitted support yields no objective or fitted parameter and terminates as invalid-input or unavailable, never as a valid zero-amplitude fit.
SCI-BEAM-PRED-013	A compatible correct soft prior may change locator work but, under the same selected basin and measurement support, does not change the prior-free measurement optimum.
SCI-BEAM-PRED-014	Broad, weak, missing, incompatible, or unsuccessful priors select the declared blind route and cannot fabricate prior success.
SCI-BEAM-PRED-015	When a wrong compatible prior conflicts with a stronger observed candidate included by the blind route, common-objective comparison selects the better data-supported candidate and records the conflict.
SCI-BEAM-PRED-016	Permuting detector rows leaves stable-identity results invariant; duplicate or ambiguous bindings are rejected rather than resolved by position.
SCI-BEAM-PRED-017	Repeated slot labels do not merge detectors when stable occurrence-scoped bindings differ.
SCI-BEAM-PRED-018	Missing or incomplete response/kernel state preserves fitted effective-profile parameters where otherwise admissible but makes an intrinsic detector-plus-telescope beam claim unavailable.
SCI-BEAM-PRED-019	With correlated noise and its correct covariance, repeated standardized parameter errors follow the declared likelihood calibration; a deliberately diagonal approximation may preserve a point estimate yet its uncertainty remains conditional and can be miscalibrated.

ID	Predicted invariant or state
SCI-BEAM-PRED-020	A sequence alternating between selected candidate identities never satisfies the separately recorded candidate-stability component and terminates with an alternating-candidate or maximum-iteration cause, not valid convergence.
SCI-BEAM-PRED-021	Optimizer completion with an unstable per-parameter metric, availability state, objective, candidate, support, or valid-detector set is non-converged. Angles straddling 0 and π are compared by d_π , angle unavailable at circularity is omitted only when unavailable at both transitions, and exhausting $N_{\max}$ records maximum-iteration termination.
SCI-BEAM-PRED-022	Repeating identical observations in different run orders yields identical observation-local results and identities apart from execution provenance; no scientific state leaks from preceding runs.
SCI-BEAM-PRED-023	Candidate-factor uncertainty changes when source-flux or amplitude uncertainty changes; if their dependence is unknown, the candidate uncertainty is unavailable rather than computed as independent.
SCI-BEAM-PRED-024	Identical numerical payloads paired with different validity, response, calibration, or authority states remain different scientific results and cannot support the same downstream claim by payload equality alone.

C Bounded references

1. S. Bryan et al., *Optical Design of the TolTEC Millimeter-wave Camera*, 2018, arXiv:1807.00097, version 1. Used only for the bounded context stated above.
2. J. Golec and the TolTEC Collaboration, *Early high-resolution millimeter-wave maps from TolTEC*, 2024, <https://doi.org/10.1051/epjconf/202429300022>. Used only for the bounded commissioning/output context stated above.